

Workshop on Developing Standards and Documentation For LLM Use as Simulated Research Participants

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1 Why This Workshop?

Alongside the recent excitement towards generative AI, there has been a surge of interest in using large language models (LLMs) as simulated research participants. Researchers and technologists have begun experimenting with LLMs to simulate various forms of human participation in research, from conducting interviews [3] to generating responses in surveys [7] or text annotation tasks [8]. Far from being a speculative exercise, LLMs are now being applied across a range of contexts within research [4, 9, 10] and industry settings [5, 12]. The motivations behind these substitution proposals are varied and include goals such as increasing the speed and scale of research, reducing costs, augmenting the diversity of collected data, or protecting participants from harm [1].

While LLMs may be championed for their pragmatic value, their use invites epistemological questions that must first be understood and addressed by the CHI community. Simulating human participants pushes the boundaries of what it means to produce knowledge in HCI, a discipline that has long prioritized contextually situated inquiries into human experiences, behaviors, and interactions with technology, particularly through the core HCI practice of user experience (UX) research. LLMs trained on vast datasets could produce 'believable' and fluent human-like responses, but they lack the embodied lived experiences that human participants bring to research [6]. What are the limits of a model's ability to produce knowledge that is rooted in specific cultural, social, and material conditions, given their disembodiment and inherently partial representation

of human experience? These questions demand a critical reevaluation of what counts as 'valid' data in HCI, the ethics of using LLM-generated data, and a reckoning with the ways that knowledge generated from simulated interactions is different from those derived from human experiences with technology.

We must also contend with the ethical implications of using LLMs as human subjects in HCI research. Most current LLMs are trained on public content extracted from the Internet [2], often without explicit consent from data subjects. This lack of informed consent may undermine the autonomy of data subjects, which is central to research ethics. In addition, the opaque nature of LLM training processes makes it difficult to trace specific pieces of data back to their authors, as much data lacks author linkage with text [13], complicating efforts to obtain retroactive consent or provide meaningful transparency.

In addition to ethical concerns, using synthetically generated research data may also lack validity. Consider how a model trained predominantly on Western-centric datasets may not accurately simulate experiences from non-Western contexts, limiting its utility in research that seeks to understand perspectives from communities in the Global South. Qualitative researchers increasingly surface concerns when using LLMs either as simulated research participants [6] or as analysis tools [11]: using LLMs in these functions can lack depth and contextual nuance, flattening the richness of human perspective and insight. Different models undergo different alignment processes depending on the goals and values of the developers. For instance, models like OpenAI's ChatGPT are aligned to avoid generating certain harmful or offensive content; on the other hand, platforms like Gab.ai which market themselves as 'uncensored and unfiltered,' or XAI's Grok, might produce outputs that are less moderated and could align more closely with conservative ideologies. Given recent work showing the risks of language generated about minority groups when intending to represent them [13] researchers should exercise caution when drawing inferences from simulated interview data.

In considering the use of LLMs as simulated research participants in HCI, it is important to acknowledge and address the complexity of determining if, when, and how they can be applied appropriately across different research contexts. HCI, as a multidisciplinary field, engages with diverse methodologies and research goals, ranging from usability testing and design exploration to critical inquiries

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into human-technology interaction. The decision to incorporate LLMs requires nuanced consideration of the specific research objectives and types of knowledge being sought. For example, in user experience research, some might find LLMs useful for pilot testing interview protocols, allowing researchers to iterate quickly and refine questions before engaging real participants. On the other hand, the appropriateness of using LLMs might be contested when the research seeks to explore deeply situated human experiences, cultural practices, or sensitive topics. Here, the absence of embodied, lived experience in LLM-generated responses can introduce risks of misrepresentation, oversimplification, or even harm, particularly when studying marginalized or vulnerable populations [6].

This workshop seeks to foster critical dialogue around the situational and context-specific considerations of when and how to employ LLMs as simulated research participants in HCI. Through a series of discussions and collaborative activities, participants will explore the benefits and limitations of using LLMs in various HCI contexts, paying particular attention to issues of validity, ethics, and equity. Our goal is to develop guidelines that address the nuances of incorporating LLMs into research, ensuring that their use aligns with the epistemological commitments and ethical standards of HCI.

This workshop will serve as the first step to develop guidelines, standards, and documentation practices for LLM use in HCI human subjects research. Workshop participants will be invited to submit reflection on suggested criteria for using LLM-generated data in UX research, which will provide the basis for a first draft of suggested guidelines, which will be discussed at the workshop. Based on engagement in the workshop, participants will join an extended collaboration as coauthors to synthesize and finalize these guidelines. After publication, we will work with conferences like CHI, educators, and standards organizations to determine if parts of our guidelines can be useful to their missions. The outputs of our workshop will be broadly applicable to not just academic research, but also industry user experience and human subject research.

2 Length of Workshop

Short (one 90 minute sessions). This length will ensure we have adequate time to introduce the topic and context to participants and hold several breakout sessions to progress writing a position paper.

3 Who's Organizing

- **William Agnew** is a CBI postdoc fellow at CMU. William received his Ph.D. from University of Washington with Sidd Srinivasa, where he worked on AI ethics, critical AI, and robotics. William also helped found Queer in AI. William is interested in developing and sharing tools and ideas that go beyond participatory design and allow marginalized individuals and communities to own and meaningfully control their data and models derived from that data. Building on ideas from usable security/privacy, usage licenses, and indigenous data sovereignty, William wants to contribute to data and AI futures where individuals and communities know where their data is and can remove, add, or change their data in different datasets.

- **Shivani Kapania** is a Ph.D. student at the Human-Computer Interaction Institute at Carnegie Mellon University. Her research focuses on creating mechanisms for accountability and greater worker agency in interactions with machine learning systems configured as part of transnational AI supply chains.
- **Hope Schroeder** is a PhD student in Media Arts and Sciences at MIT. Her research focuses on language technologies and social research methods, including those for analyzing small group dialogue and new survey designs.
- **Marianne Aubin Le Qu  r  ** is a PhD Candidate in Information Science at Cornell Tech. Her research focuses on how AI and other emerging technologies impact news and civic information consumption, including new methods for using AI to analyze text data.
- **Sarah Fox** is an Assistant Professor at Carnegie Mellon University in the Human-Computer Interaction Institute, where she directs the Tech Solidarity Lab. Her research focuses on how technological artifacts challenge or propagate social exclusions by examining existing systems and building alternatives.
- **Hoda Heidari** is the K&L Gates Career Development Assistant Professor in Ethics and Computational Technologies at Carnegie Mellon University, with joint appointments in Machine Learning and Societal Computing. Her research is broadly concerned with the social, ethical, and economic implications of Artificial Intelligence, particularly issues of fairness and accountability through the use of Machine Learning in socially consequential domains.

4 Publishing Plans

After the CHI workshop, we plan to produce a report of the workshop outcomes. The workshop papers and results will be available on the workshop website prior to and after the workshop, providing opportunities for a larger audience to get familiar with this area. We would consider publishing a collection of the submitted papers as workshop proceedings (e.g., via ArXiv). In addition, we may seek opportunities for an edited book or a special issue in a selected journal (e.g., ToCHI) where the participants will be encouraged to publish their work.

A central goal of this workshop is community building for researchers and practitioners in this area—spurring conversation between those engineering and using the cutting edge of LLM-generated research subjects, as well as those who may be more critical. After the workshop, we plan to create a platform for community members to continue the discussion and share resources. Potential options may include a periodical email newsletter, a public GitHub repository, or a Slack/Discord channel. Participants and organizers will discuss the next steps at the workshop.

5 Accessibility

To make our workshop inclusive and accessible to all participants, we will require descriptive alt-text in all submission materials, including papers and presentation slides, as well as the workshop outputs. For live presentations, real-time captions will be provided to ensure accessibility. We will also ensure that our event website

is designed with accessibility in mind, including compatibility with screen readers and other assistive technologies. During the workshop, organizers will provide technical support to presenters and participants, and we will actively seek feedback for continuous improvement. We would deeply appreciate transcription services during our workshop.

6 Offline Materials

We will post accepted papers on our website before the workshop. We will also post our draft standards and documentation practices for LLM use before the workshop, along with discussion prompts. We will create a mailing list to coordinate future workshops and discussions.

7 Activities

Participants will be asked to submit short position papers on standards and documentation for LLM use in UX research. Workshop organizers will synthesize these papers into a pre-workshop reading highlighting areas of consensus and contention. Our workshop will feature two breakout sessions to discuss and build on these positions, focusing on answering two critical questions for this set of methods: *When Should LLMs be Used for UX research?* and *How should LLM use in UX research be Documented and Validated?*

- 0:00-0:15: (all times local) Introduction – Workshop Organizers
- 0:15-0:30: Breakout 1: When Should LLMs be Used for UX research?
- 0:30-0:40: Group Discussion 1: When Should LLMs be Used for UX research?
- 0:40-1:05: Breakout 2: How should LLM use in UX research be Documented and Validated?
- 1:05-1:15: Group Discussion 2: How should LLM use in UX research be Documented and Validated?
- 1:15-1:25: Next Steps and Writing Assignments
- 1:25-1:30: Conclusion

8 After the Workshop

The purpose of this workshop is to co-create standards and documentation practices of LLM use in human subjects research for HCI researchers. After our CHI workshop, we will invite attendees to additional co-working sessions and host a virtual workshop to present a final draft of our standards. Following the creation of our standards, we will submit a jointly authored paper to a relevant venue and work with HCI research bodies (e.g., SIGCHI) to identify how our standards and documentation practices could help improve the quality and transparency of research using LLMs to replace humans.

9 Call for Participation

The Workshop on Developing Standards and Documentation For LLM Use in HCI Human Subjects Research aims to bring the HCI community together to develop standards, guidance, and documentation for the use of large language models (LLMs) as simulated research participants. This includes using LLMs in place of humans during interviews, surveys, focus groups, usability testing, co-design, and other UX research methods, but also use of LLMs in

place of human interaction upstream of studies, including use of LLMs to train for interviews, test survey instruments, or generate directions for research on people. These methods have garnered wide attention and are seeing increasing use within HCI and other communities. However, they come with many potential risks and issues, including bias, reproducibility challenges, disempowering studied populations, and parroting researcher positions.

We invite participants to submit extended abstracts and position papers of up to four pages (excluding references) on LLM use in UX research. We encourage participants to address some or all of the following questions in their position papers:

- How should we taxonomize LLM use in UX research?
- When should we use LLMs in UX research?
- How should we document LLM use in UX research?
- How should we validate LLM use in UX research?

Workshop organizers will make accepted position papers, and a summary synthesizing areas of agreement and contention, available to participants prior to the workshop. After the workshop, participants will be invited to collaborate on a position paper on standards for LLM in UX research use, documentation, and validation.

Workshop attendees will also be invited to collaborate on a joint paper proposing standards and documentation for LLM use in UX research.

10 Expected size of attendance

We expect approximately 50 attendees.

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